

Article

Fiscal Pressure and Public–Private Partnership Investment: Based on Evidence from Prefecture-Level Cities in China

Junjiao Gong ^{1,2,*}, Yingyu Lu ¹, Yang Xu ³ and Jincun Fu ⁴

¹ College of Business, Wenzhou University, Wenzhou 325035, China

² Wenzhou Economic Research Institute, Wenzhou University, Wenzhou 325035, China

³ Inspection and Maintenance Department, Sinopec South Chemical Company, Nanjing 210096, China

⁴ College of Business, North Minzu University, Yinchuan 750030, China

* Correspondence: 00011072@wzu.edu.cn; Tel.: +86-138-0688-3628

Abstract: Public–private partnership (PPP) policy is essential to alleviating the local government debt burden and improving resource allocation efficiency. This paper empirically examines the impact of fiscal pressure on PPP investment in Chinese prefecture-level cities from 2014 to 2019 using the ordinary least-squares (OLS) module. Moreover, we also investigate how fiscal pressure influences PPP investment and test the influenced mechanism from other perspectives. The results show the following. (1) Fiscal pressure on the government has a significant positive effect on PPP investment at the prefectural level. (2) The marketization process is the mediated effect of the relationship between fiscal pressure and PPP investment. Fiscal pressure will stimulate the regional marketization process, thus promoting PPP investment. (3) Fiscal pressure has a significant positive effect on PPP investment in the middle region, while the effect is not significant in the eastern and western regions. Meanwhile, the effect is not significant in central cities, but there is a significant positive effect in ordinary cities. (4) The effect of fiscal pressure on PPP investment is not significant in the private reward mode of government payment, but there is a significant positive effect in the mode of user payment and feasibly insufficient subsidies. Our studies could also provide practical suggestions for sustainable development of PPP policy and solving the fiscal pressure of the current economic recession under the COVID-19 pandemic.

Keywords: fiscal pressure; PPP investment; marketization process



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1. Introduction

Public–private partnership projects (PPPs) are necessary to ensure infrastructure construction for sustainable development in the current economic downturn due to the COVID-19 pandemic outbreak, constituting a mode of cooperation between the public and private sectors, whereby the public section transfers to private sector entities a degree of responsibility for the execution of public policy projects. PPP projects have been shown to develop better under the circumstances of stable macroeconomics, sophisticated legal systems, and higher levels of openness to the outside world (Hammami et al., 2006) [1]. With the rapid development of urbanization, the demand for infrastructure related to livelihood is gradually increasing. Moreover, if the infrastructure relies only on the government's financial investment, it will trigger the dissatisfaction of social demand both in terms of the quantity and quality of products, resulting in an increase in the government's fiscal deficit and debt crisis. Therefore, as the government needs to promote infrastructure and industry, it should give its full attention to establishing PPP policy as a means of alleviating the fiscal pressure and stabilizing local investment by reasonably defining the boundaries of their own authority, reducing the ex post default, and actively encouraging the participation of private capital (Gong et al., 2019) [2].

In China, according to the 2021 statistics from the Center for Government and Social Capital Cooperation, the number of PPPs totaled 10,034 with a total investment of

CNY 1.55 billion. The PPP investment mode is a long-term partnership established for infrastructure construction and public services offering advantages to the public and private sectors and improving the efficiency and quality of public goods supply. Some research showed that PPP policy could relieve fiscal pressure via deferred payment for public services and products (Shen et al., 2020; Yuan et al., 2017) [3,4]. China went through a tax-sharing reform in 1994 and an income-tax-revenue-sharing reform in 2002, respectively. This is the reason why PPP policy developed so rapidly in China. With the PPP mode, the local government can decentralize payment on basic design and public services, which greatly relieves the pressure of excessive expenditure.

This paper will explore the influential factors of PPP investment from the perspective of fiscal pressure and further examine how the relationship between fiscal pressure and PPP investment can be influenced. A few papers discussed how the development of the marketization process can force the government to carry out institutional reforms and reduce the inefficiency of cooperation due to information asymmetry (Shen et al., 2020, Feng et al., 2020) [3,5]. Most of the prior literature takes the latter perspective, mainly discussing the effect of the private sector participant, while we stress the role of the public sector participants in PPP projects as academic motivation for the paper. In particular, this paper implicitly assumes that public sector actors are rational players, motivated to maximize their own wealth utility (Shleifer, 1998) [6]. Furthermore, the marketization process can encourage private sector participants to participate in PPP projects, and with the development of marketization, it can force the government to carry out institutional reform and innovation to improve the operational efficiency of PPP projects.

This study provides a few contributions. First, we discuss the relationship between fiscal pressure and PPP investment, which not only enriches the research on PPP policy but also provides a new possible solution to the local government's financing dilemma. Second, this paper explores how fiscal pressure influences PPP investment, being the first study to discuss the mediated effect of the marketization process, providing a point of reference for policy committees and informing regional regulations. Third, our research also provides practical contributions, encouraging governments to make PPP policy a component in promoting regional sustainable development. We adopt a practical perspective, emphasizing PPP policy's ability to reduce fiscal pressure, as demonstrated by China's actual conditions. Meanwhile, we also discuss the effect in various regions and provide novel insights into studies on regional fiscal pressure and PPP policy. Fourth, we further discuss the relationship between fiscal pressure and PPP investment in the different modes of PPP projects, which is crucial for formulating differentiated measures of solving fiscal pressure. The results of this empirical study will therefore be valuable for local governments seeking to renew the existing policy in light of current suggestions.

The rest of the paper is organized as follows: Section 2 comprises the literature review and hypotheses; Section 3 presents the research design; Section 4 provides the conclusion and analysis; Section 5 is the verification of the mechanism; Section 6 summarizes our findings and their implications.

2. Literature Review and Hypothesis

2.1. Literature Review

2.1.1. The Effect of Fiscal Pressure on PPP Investment

This research explored the connection between fiscal pressure and PPP investment in China. This motivated us to focus on a review of the available literature on the factors influencing PPP investment. The investment in PPP projects had a counter-cyclical impact on quantitative GDP increases during the recession period, and the results show that PPP investment can improve the quality of regional economic growth by enhancing innovation and entrepreneurship at the regional levels (Wang et al., 2022) [7]. The previous literature shows evidence for three influencing factors, including economics, institution, and government, affecting PPP project development and investment. General determinants, applicable to any jurisdiction, follow. One determinant would be the government's ideo-

logical orientation regarding the relevant policy issue. This refers to the ideological mix (*laissez-faire* economics versus the developmental state) underpinning the government's approach with respect to the particular dimension of economic policy (Gomez, 2009) [8]. Geographical region may be another determinant; Ma and Li (2019) [9] also discovered that the success of the PPP projects in the eastern region is much higher than that in the central and western regions through a regional heterogeneity test, which indicates that the economic environment has a positive effect on the success of PPP projects. Another determinant would be the level of institutional sophistication within the country. Luo et al. (2017) [10] showed that institutional quality affects the effectiveness of PPP projects by influencing the degree of risk sharing of the private sector in PPP projects, and the results were obtained from an empirical study on PPP projects in 46 developing countries along the Belt and Road. Feng et al. (2017) [11] studied 28 typical cases of PPP projects in China and found that the institutional environment and system for PPP projects are key factors affecting their success. Casdy (2019) [12] explored the impact of the institutional environment on the maturity of the PPP project market in terms of legitimacy, capacity, and trust, and concluded that sophisticated institutions and governance would contribute to the sustained growth of PPP projects in the infrastructure construction sector, thus promoting the development of a mature PPP project market. Jiang et al. (2021) [13] found that the institutional quality of government has a positive effect on PPP investment in infrastructure using panel data. A third determinant may be the government's level of fiscal resources. Opara et al. (2017) [14] found that the prerequisites for the success of PPP projects are factors, strong government leadership support, a favorable policy environment, and effective government organizational capacity. The geographical region may be another determinant, for a reason specific to the PRC. There are disparities in the level of economic development and degree of government intervention across different geographical regions within this country (Yu and Wang, 2020) [15]. One reason why policymakers opt for PPP includes the need of the government to tap into the private sector's skill and expertise, the necessity of the public sector to concentrate on policymaking, and the urgent need to meet the increase in demand for infrastructure (Enebeli, 2022) [16]. Zhang and Zheng (2015) [17] found that the higher the level of regional economic development, the higher the number of PPP projects issued.

Although PPP projects have been adopted more extensively in larger numbers and with high investments, there is still a small amount of research on the impact of fiscal pressure on PPP investment, and the findings are inconsistent. In general, governments are more inclined to adopt the PPP policy when they are in fiscal distress. Buso et al. (2017) [18] suggested that the PPP policy can better address the problem of moral hazards compared to traditional government procurement models when governments face fiscal constraints, which holds true when both governments and social capital face fiscal constraints. Albalade et al. (2014) [19] found that fiscal pressure had a positive effect on the private capital proportion in PPP projects. However, some of the related literature concluded a negative relationship between fiscal pressure and PPP policy. Yuan et al. (2017) [4] found a significant negative relationship between the local government's financial resources and PPP projects relating to transportation facilities using data at the county and city levels in China. In addition, Castelblanco et al. (2022) [20] explored recovery measures to address short- and long-term PPP-road-project-related challenges associated with the COVID-19 pandemic by considering recuperation actions implemented during the 2008 global financial crisis (GFC). Shen et al. (2020) [3] indicated that there is an inverted U-shaped relationship between fiscal pressure and PPP projects, and found that PPP policy could relieve fiscal pressure by deferred payment for public services and products; thus, the local government's financial resources have a positive effect on PPP investment. However, if the government's financial resources are scarce, financial resources will have a negative resistance effect on PPP investment, as scarce financial resources make private companies doubt the government's affordability, and thus reduce the motivation of private capitalists.

2.1.2. The Effect of the Marketization Process on the Relationship between Fiscal Pressure and PPP Investment

PPP projects not related to hard infrastructure encounter challenges embodied by the unclearness of the value creation mechanism. From the perspective of the partnership attribute, this paper puts forward two variables, market embeddedness (partner attribute) and project complexity (content attribute), to develop a theoretical framework for a value creation mechanism in public culture PPP projects (Yingying et al., 2022) [21]. The PPP policy is a mode of public–private partnership, in which the government signs a contract with private companies to hand over part of the construction and operation rights of infrastructure construction. However, in practice, the proportion of private capital is low in PPP projects in China, being mostly maintained at about 35% (Shen et al., 2020) [3]. Hence, it is an urgent issue that needs to be addressed to increase the proportion of private capital. Some studies have shown that if fiscal pressure is low, a sound business environment can alleviate the financial pressure and decline in investment in PPP projects. Moreover, Jia et al. (2022) [22] explores the evolutionary trend of social capitals' cooperative behavior under different influences. It was found the default penalty is positively related to the evolution of social capitals' cooperative behavior. Additionally, the development of the marketization process can force the government to carry out institutional reforms and reduce the inefficiency of cooperation due to information asymmetry. Furthermore, the advancement of marketization can influence the change in government perception, policy, and regulation making (Gong et al., 2020) [23]. Le et al. (2022) [24] studied the risk catalog and showed that the most frequent risks demonstrated in the literature are "low traffic demand", "escalation of operation-maintenance cost", "inflation rate fluctuation", "problems with land acquisition and compensation", "change in laws and regulations" and "interest rate fluctuation". Under financial pressure, the government could relinquish their right to control, which is conducive to the government focusing on project governance, service provision, and support and assurance. Additionally, it can help entrepreneurs to reduce costs and increase efficiency, thus helping to improve cooperation efficiency, promote enterprise innovation, and stimulate market vitality (Feng et al., 2020) [5]. What is more, for effective implementation of PPP road projects in Ethiopia, the government should at least ensure the presence of an enabling specific PPP policy, well-organized and committed public agency, a stable political and social environment, favorable legal frameworks, and good governance (Debela, 2022) [25]. The evolution of the body of knowledge on PPP and the relationships between its main components has been analyzed over the last two decades. Results demonstrate that the body of knowledge on PPP has been traditionally driven by the economic dimension of sustainability, neglecting the social and environmental dimensions, which should be strengthened to counterbalance the current pitfalls in social legitimacy and environmental underperformance to promote the emergence of more sustainable PPPs (Castelblanco, 2022) [26].

2.2. Theory and Hypothesis

2.2.1. The Impact of Fiscal Pressure on PPP Investment

In 1994, China went through a tax-sharing reform, which shifted the financial rights of local governments upward, thus reducing its financial revenue, while the financial expenditures of local governments remained. This led to the emergence of the problem of imbalance between local financial revenues and expenditures. The study demonstrates the most critical sustainable performance criteria of PPP projects, such as the lowest project costs, green index, disability-friendly inclusion rate, and carbon emission per project, among others (Akomea et al., 2022) [27]. As the urbanization process continues to advance, the demand for infrastructure and public services increases dramatically. Local governments carry out projects in the public sector through local financing, which accumulates a large amount of hidden debt risk due to the imperfect construction of local debt-related laws and regulations. There is a significant positive association between public–private partnership investment in the energy and ecological footprint in the long

run and the short run, and an increase in public–private partnership investment in energy was shown to affect the environmental sustainability of Pakistan (Li et al., 2021) [28]. Furthermore, to prevent the outbreak of hidden debt risk for local governments, the State Council promulgated the Opinions of the State Council on Strengthening the Management of Local Government Debt in 2014, emphasizing the scale control, budget management, prevention, and resolution of local government debt risk. Meanwhile, the Ministry of Finance issued the Notice on Issues Related to the Promotion and Application of the Government and Social Capital Cooperation Model, and the PPP mode developed rapidly in China. Adebayo et al. (2021) [29] utilized the frequency domain causality test to capture the causal impact factors of PPP investment in energy, renewable energy consumption, technological innovation, and economic growth. Moreover, the PPP policy was developed to reduce fiscal pressure and alleviate the risk of debt in the process of implementation. In recent years, PPP policy has permeated into infrastructure construction and public services, greatly alleviating the financing pressure on local governments. Therefore, we proposed the following hypothesis:

Hypothesis 1 (H1). *Local fiscal pressure has a positive effect on PPP investment.*

2.2.2. The Mediating Effect of the Marketization Process in the Relationship between Fiscal Pressure and PPP Investment

To alleviate project financial pressure and improve performance, the public–private partnership (PPP) arrangement was introduced by the central government of China to facilitate the sustainable development of infrastructure (Zhou et al., 2022) [30]. Although PPP policy is developing rapidly in China, the proportion of private capital is low. Furthermore, most PPP projects are conducted by state-owned enterprises and central state-owned enterprises with local governments, which dismisses the advantage of financing marketization. In general, the government and private companies sign a particular contract to protect the rights and interests of both sides. PPP investment can significantly improve the quality of regional economic growth, and PPP investment can improve the quality of regional economic growth by enhancing innovation and entrepreneurship at the regional levels (Wang et al. 2022) [7]. However, the government prefers to use its authority to force private capital to bear the public burden. The government also frequently changes policies and regulations, so there is great uncertainty about the revenues of private capital, which influences the enthusiasm for private capital participation. Public–private partnerships (PPPs) are identified as potentially beneficial procurement strategies which boost capacities to manage risks by pooling diverse resources (Jayasena et al. 2022) [31]. The Third Plenary Session of the 18th Central Committee declared “let the market play a decisive role in the allocation of resources”. The development of marketization is conducive not only to the government’s simplification and decentralization but also to the government’s reform and innovation in the system. Private investors may have a greater understanding and more information about the market than governments when they propose a PPP project (Kim et al., 2021) [32]. Consequently, if the fiscal pressure is large, the government prefers to promote private capital participation, resulting in stimulating market dynamics. Sequentially, the marketization process develops dynamically, and private capital is active in participating in PPP policy, resulting in stimulating the development of PPP projects. Specifically, we hypothesize the following:

Hypothesis 2 (H2). *The marketization process is the mediated effect of the relationship between fiscal pressure and PPP investment.*

3. Methodology

3.1. Model

This study examined the impact of fiscal pressure on PPP investment, so that the amount of PPP investment in prefecture-level cities is the dependent variable and fiscal

pressure is the core independent variable in our model. However, there are many factors affecting PPP project investment in prefecture-level cities, and if only the above two variables are included in the model, the estimation results will generate serious bias. For this reason, we needed to further select the factors affecting PPP project investment in prefecture-level cities as control variables to be introduced into the model for removing the influence of these factors, and then analyze the relationship between fiscal pressure and PPP project investment in prefecture-level cities. We transformed all parameters into a natural log for empirical examination to use a log-linear formation instead of a linear formation. Shahbaz et al. (2012) [33] concluded that a log-linear form offers much more constant and accurate outcomes. Based on these claims, to estimate the hypotheses, we established the model as follows:

$$\ln PPP_{it} = \beta_0 + \beta_1 deficit_{it} + \beta_2 X_{it} + \theta_t + \varepsilon_{it} \quad (1)$$

where PPP_{it} is the dependent variable, which denotes the amount of PPP investment in the i prefecture-level city for t year. $deficit_{it}$ is the main independent variable, and represents the fiscal pressure in the i prefecture-level city for t year. X_{it} is the control variable, and represents the factors affecting PPP investment in prefecture-level cities, including the urbanization rate, the inflation rate, the secondary industry share, per capita GDP, resident population, and the fixed asset ratio. In addition, θ_t represents fixed effects, and ε_{it} is the random error. All variables in our paper are clearly defined in Table 1.

Table 1. Variables' Definition.

Variables	Variables' Definition
$\ln PPP$	Logarithmic value of amount invested in PPP projects at the prefectural level
$deficit$	Local government fiscal pressure, $(EX - RE)/GDP$
$urban$	Urbanization rate, UP/TP
$infla$	Inflation rate, $(CP - BP)/BP$
$indus$	Share of secondary industry, OV/GDP
$pergdp$	Per capita GDP, GDP/TP
$\ln people$	Logarithm of resident population
$\ln asset$	Fixed assets/GDP

Note: where *expend* denotes the general public budget expenditure of prefecture-level municipal governments, *revenue* denotes the general public budget revenue of prefecture-level municipal governments, and GDP denotes the gross domestic product of prefecture-level municipalities. UP denotes the urban population, and TP denotes the total population in the city. CP denotes current price level, and BP denotes base period price level. OV denotes output value of secondary industry.

3.2. Data Source

In 2014, the State Council in China systematically elaborated on the PPP mode (these are explored in the Guidance Opinions of the State Council on Innovation of Investment and Financing Mechanism in Key Areas to Encourage Social Investment), thus laying the foundation for the development of the PPP mode. Referring to the previous literature (Yuan, 2017; Zhou, 2019; Wang, 2019; Buso, 2019) [4,34–36] considering the availability of data, we selected the data on PPP projects of prefecture-level cities ($\ln PPP$) in China from 2014 to 2019 using whole-group sampling, and the data are from the Center for Government, Social Capital Cooperation of the Ministry of Finance, and the WIND database (the WIND database is a provider of financial data and analytical tools, and Information is a mainland China financial data, information and software services company headquartered in Shanghai's Lujiazui Financial Center. <https://www.wind.com.cn/portal/en/Home/index.html>, accessed on 23 January 2021). We also collected data on economics at the prefectural level, such as regional urbanization rate (*urban*), inflation level (*infla*), secondary industry share (*indus*), per capita GDP (*pergdp*), resident population (*ln people*), and population fixed asset investment (*ln asset*) from the China Urban Statistical Yearbook, the National Economic, and Social Development Bulletin. The descriptive statistics of various variables are shown in Table 2. The maximum value of the natural logarithm of the investment amount of

PPP investments in prefecture-level cities is 16.69 and the minimum value is 6.701, which indicates that there are large differences in the investment scale of PPP projects between different cities. And the maximum value of the financial pressure on local governments is 2.831 and the minimum value is -0.382 , which expresses that there are large differences in the financial pressure between different prefecture-level cities. The economic characteristics of prefecture-level cities were considered, such as the mean value of the urbanization rate, 0.569, with a fluctuation range of 0.226 to 1, which shows that the urbanization level varies greatly among different prefecture-level cities. Moreover, the mean value of the inflation rate is 1.019, with a fluctuation range of 0.997 to 1.042. The mean value of the proportion of the secondary industry is 0.444, with a fluctuation range of 0.114 to 0.882. The mean value of per capita GDP is 5.521, with a fluctuation range of 1.047 to 21.549. The mean value of the natural logarithm of the resident population of prefecture-level cities is 5.903, with a fluctuation range of 3.171 to 8.055. Finally, the mean value of the natural logarithm of fixed asset investment is -0.23 , with a fluctuation range of -2.848 to 2.427. The results show that the investment scale of PPP projects, financial pressure, and urbanization level had significant differences among different cities. This motivated the inclusion of these variables in the empirical model as potential covariate determinants of the level of PPP at the prefecture level.

Table 2. Descriptive statistics of variables.

Variables	Mean	SD	Min.	Max.	Sample Size
<i>lnPPP</i>	12.799	1.5	6.701	16.69	1746
<i>deficit</i>	0.147	0.149	-0.382	2.813	1746
<i>urban</i>	0.569	0.133	0.266	1	1746
<i>infla</i>	1.019	0.007	0.99	1.042	1746
<i>indus</i>	0.444	0.107	0.114	0.882	1746
<i>pergdp</i>	5.521	3.317	1.047	21.549	1746
<i>lnpeople</i>	5.903	0.679	3.171	8.055	1746
<i>lnasset</i>	-0.23	0.491	-2.848	2.427	1746

Note: standard errors at the prefectural level are in parentheses.

4. Results and Analysis

4.1. Estimated Results

We used the fixed-effects model (For this study, the fixed-effects model outperformed the random-effects model by the Hausman test.) to estimate the influence of fiscal pressure on PPP investment at the prefectural level. Before the estimation, we conducted the correlation analysis, the results of which are presented in Table 3, showing that the relationship between local government fiscal pressure and PPP investment is significantly correlated, tentatively supporting hypothesis 1 (H1). Additionally, Table 4 shows the estimated results using the stepwise regression method to solve the problem of multicollinearity from Column (1) to Column (7). Additionally, Column (7) shows that the fiscal pressure on prefecture-level municipalities positively significantly influences the investment in PPP projects at the level of 1%. This means that the high fiscal pressure on prefecture-level municipalities could stimulate the PPP investment, verifying hypothesis H1. Since the tax-sharing reform in China in 1994, an upward shift in local fiscal power and reduced local fiscal revenue has been observed. To prevent the over-borrowing behavior of local governments, the PPP investment mode was introduced into the public service sector to alleviate the fiscal pressure and resolve the debt risk. Therefore, the high fiscal pressure on local governments led to the implementation of the PPP investment mode. In addition, the urbanization rate positively influences PPP investment at the 5% significance level. With the great mitigation from rural areas to cities, the level of urbanization has been increasing, resulting in a greater demand for urban infrastructure construction. Consequently, local governments are looking to promote the development of infrastructure PPP projects.

Table 3. The correlation analysis.

Variables	<i>lnPPP</i>	<i>deficit</i>	<i>urban</i>	<i>infla</i>	<i>indus</i>	<i>pergdp</i>	<i>lnpeople</i>	<i>lnasset</i>
<i>lnPPP</i>	1							
<i>deficit</i>	0.152 ***	1						
<i>urban</i>	0.135 ***	−0.437 ***	1					
<i>infla</i>	−0.126 ***	−0.104 ***	0.116 ***	1				
<i>indus</i>	0.046	−0.373 ***	0.154 ***	−0.210 ***	1			
<i>pergdp</i>	0.159 ***	−0.451 ***	0.804 ***	0.214 ***	0.220 ***	1		
<i>lnpeople</i>	0.356 ***	−0.358 ***	0.126 ***	0.119 ***	−0.105 ***	0.198 ***	1	
<i>lnasset</i>	0.050 *	0.210 ***	−0.420 ***	−0.069 **	0.100 ***	−0.284 ***	−0.055 **	1

Note: Standard errors at the prefectural level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4. The estimation results of H1.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>deficit</i>	1.341 *** (0.512)	1.888 *** (0.575)	1.756 *** (0.581)	1.841 *** (0.574)	1.849 *** (0.565)	1.846 *** (0.544)	2.741 *** (0.948)
<i>urban</i>		−1.323 (1.968)	9.001 *** (3.336)	7.892 ** (3.931)	8.992 * (5.241)	12.98 *** (4.920)	10.99 ** (4.991)
<i>infla</i>			−0.630 *** (0.106)	−0.619 *** (0.105)	−0.600 *** (0.111)	−0.620 *** (0.112)	−0.632 *** (0.117)
<i>indus</i>				−1.081 (1.897)	−1.155 (1.918)	−0.616 (1.796)	−0.522 (1.828)
<i>pergdp</i>					−0.00690 (0.0147)	−0.0157 (0.0155)	−0.0133 (0.0157)
<i>lnpeople</i>						3.003 (2.044)	2.533 (2.067)
<i>lnasset</i>							−0.357 (0.273)
Constants	12.62 *** (0.0703)	13.48 *** (1.118)	71.85 *** (9.819)	71.75 *** (10.50)	69.62 *** (11.52)	51.60 *** (17.43)	56.41 *** (17.56)
City fixed effect	Control	Control	Control	Control	Control	Control	Control
Observations	1746	1746	1746	1746	1746	1746	1746
R ²	0.003	0.007	0.081	0.078	0.079	0.083	0.087

Note: standard errors at the prefectural level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Although our paper has verified the positive effects of fiscal pressure on PPP investment, we still wish to figure out in what way fiscal pressure specifically promotes PPP investment, which will help us to understand the relationship much better. PPP policy is a public–private partnership mode in which the government and private companies sign a contract to hand over part of infrastructure construction to private companies for construction or operation. The main partnership modes are BOT (Build–Operate–Transfer), BOO (Build–Own–Operate), and TOT (Transfer–Operate–Transfer). If the government's financial pressure increases, it can encourage private sector participation and stimulate market dynamics by conceding power (Zhu et al., 2020) [37]. Hence, as the fiscal pressure increases, local governments prefer to encourage private capital participation in infrastructure construction, which stimulates market dynamics and improves project operational efficiency, thus promoting PPP projects. In other words, the marketization process is the mediated effect of the relationship between fiscal pressure and PPP investment. To verify the proposed hypothesis, we established the following basic model:

$$MI_{it} = c_0 + c_1 deficit_{it} + c_2 X_{it} + \theta_t + \varepsilon_{it} \quad (2)$$

$$\ln PPP_{it} = d_0 + d_1 \text{deficit}_{it} + d_2 MI_{it} + d_3 X_{it} + \theta_t + \varepsilon_{it} \quad (3)$$

where MI_{it} denotes the degree of marketization of the i th prefecture-level city in year t . MI can be defined as the China Marketization Index, edited by Fan Gang, Wang Xiaolu, and Zhu Hengpeng. It consists of five indices: the relationship between the government and the market, the development of the non-state economy, the development of product markets, the development of factor markets, the development of market intermediary organizations, and the legal environment for maintaining markets. Additionally, it reflects the development of marketization among regions in China.

For this estimation, we estimated the relationship between PPP investment and fiscal pressure in Equation (1). Next, we estimated the effect of fiscal pressure on the marketization process in Equation (2). Finally, we estimated the mediated effect of the marketization process in Equation (3), and the results are shown in Column (3). Moreover, the data for marketization were obtained from the China Marketization Index, compiled by Fan Gang et al., and the empirical results are shown in Table 5. Specifically, Column (1) reports the results of the basic model, and the results in Column (2) indicate that there is a significant positive effect of fiscal pressure on the marketization process in prefecture-level municipalities, and Column (3) exhibits that the marketization process has a significant positive effect on PPP investment. This means that fiscal pressure will stimulate the regional marketization process, thus promoting PPP investment. This conclusion supports H2.

Table 5. The estimation results of H2.

Variables	lnPPP	MI	lnPPP
	(1)	(2)	(3)
<i>deficit</i>	2.741 *** (0.948)	0.982 *** (0.306)	2.441 *** (0.919)
<i>MI_{it}</i>			0.350 ** (0.157)
Control variables	Y	Y	Y
City fixed effect	Y	Y	Y
Observations	1746	1746	1746
R ²	0.087	0.798	0.096

Note: standard errors at the prefectural level are in parentheses. ** $p < 0.05$, *** $p < 0.01$.

4.2. Robustness Test

We conducted robustness tests using the method of replacing the core dependent variables. Referring to the literature (Ma and Li; 2019 [9]), we replaced the amount of PPP project initiation investment with the number of PPP projects. This is because a rise in investment can trigger an increase in the number of PPP projects. We used the number of PPP projects rather than the natural log of the number of PPP projects in the estimated model, since the logarithm was taken to reduce the absolute value between the data, while the number of PPP projects did not change much in our research. The results in Table 6 show that the relationship between fiscal pressure and PPP investment at the prefectural level remained significant after adding a series of control variables. Hence, the conclusion of this research is solid.

Table 6. Robustness test (the dependent variable is the number of PPP projects).

Variables	NumPPP (1)	NumPPP (2)
<i>deficit</i>	3.195 * (1.907)	6.284 ** (2.997)
<i>urban</i>		68.69 *** (16.07)
<i>infla</i>		−321.0 *** (45.57)
<i>indus</i>		3.426 (5.687)
<i>pergdp</i>		1.326 (2.896)
<i>lnpeople</i>		15.76 ** (6.696)
<i>lnasset</i>		−0.292 (0.841)
Constants	4.978 *** (0.280)	198.5 *** (52.53)
City fixed effect	Control	Control
Observations	1746	1746
R ²	0.001	0.051

Note: standard errors at the prefectural level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5. Mechanism Test

We studied the effect of the national average, which is a complex case, on the results. We addressed the problem by examining the regional mechanism and reward mechanism of PPP projects.

5.1. Regional Mechanism Test

We rearranged the sample on the basis of geographical locations, including eastern, central, and western regions, based on the original data. From Columns (1) to (3) in Table 7, which report the results, we find that fiscal pressure has a significant positive effect on PPP investment in the middle region, while the effect is not significant in the eastern and western regions. This is because the fiscal pressure in the middle regions is larger than that in other regions; PPP projects are mainly implemented in the middle regions to relieve the fiscal pressure. In the western regions, the high cost of infrastructure construction and the weakness of local government make it less attractive to social capital, thus restricting the development of PPP projects. Moreover, the number of western cities with financial expenditure for PPP projects exceeding 10% of the criterion in 2019–2033 is significantly higher than that of the other regions. Therefore, the number of PPP projects would not increase in western regions, even if the western regions had high fiscal pressures. In the eastern regions, urbanization and infrastructure have developed rapidly; thus, there is little chance for the development of PPP policy. In addition, compared with the middle and western regions, the economic development in the eastern region is faster, leading to the relatively low financial pressure on local governments, so the impact of fiscal pressure on PPP investment in the eastern region is also not significant.

Table 7. Regional Mechanism Test.

Variables	Eastern Regions	Middle Regions	Western Regions	Central Cities	Ordinary Cities
	(1)	(2)	(3)	(4)	(5)
<i>deficit</i>	1.373 (0.961)	23.23 *** (6.107)	0.832 (2.058)	−18.94 (23.42)	2.016 ** (0.933)
<i>urban</i>	34.00 *** (9.467)	9.680 (6.670)	−13.00 (9.033)	−17.13 (18.31)	12.23 ** (4.936)
<i>infla</i>	−100.2 *** (23.26)	−69.32 *** (19.87)	−26.62 (18.35)	−85.86 ** (34.74)	−61.14 *** (12.33)
<i>indus</i>	−2.042 (4.639)	0.712 (1.750)	0.664 (3.669)	−5.939 (6.915)	0.0701 (1.745)
<i>pergdp</i>	−0.324 * (0.164)	0.0192 (0.245)	−0.330 (0.275)	−0.0879 (0.246)	−0.142 (0.181)
<i>lnpeople</i>	−1.277 (7.257)	1.956 (2.539)	14.17 *** (4.795)	14.61 (8.319)	2.378 (2.172)
<i>lnasset</i>	0.0113 (0.355)	−1.053 *** (0.353)	0.102 (0.916)	−1.487 *** (0.341)	−0.0936 (0.292)
Constants	103.9 ** (44.50)	63.53 ** (28.45)	−34.85 (28.92)	17.98 (45.36)	54.85 *** (17.65)
City fixed effect	Control	Control	Control	Control	Control
Observations	600	630	516	162	1584
R ²	0.215	0.146	0.115	0.221	0.087

Note: standard errors at the prefectural level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Furthermore, we divided the cities into two categories, central cities and ordinary cities (Central cities include provincial capitals and municipalities, and the remainder are ordinary cities.). In Table 7, columns (4) and (5) show that the effect of fiscal pressure on PPP investment is not significant in central cities, but there is a significant positive effect in ordinary cities. The reason for this is that central cities have good economic development and high fiscal revenues and do not need to carry out PPP projects to relieve the fiscal pressure on municipalities at the prefectural level, while for ordinary cities, limited economic development requires the use of PPP projects to relieve the fiscal pressure on municipalities at the prefectural level.

5.2. Reward Mechanism Test

In PPP projects, the private company has three main ways of gaining revenue: government payment, feasibly insufficient subsidy, and user payment. Additionally, the government's fiscal expenditure decreases in order for the three private reward mechanisms. Therefore, we find that there were different conclusions for PPP projects with different reward mechanisms based on the results in the basic model. If the fiscal pressure augments, the PPP projects related to user payment would be greatly impacted, followed by those related to feasibly insufficient subsidies, and the least affected are those related to government payment. Consequently, our paper rearranged the data in line with three private reward mechanisms and estimated the basic model. Table 8 reports the results; we found that the effect of fiscal pressure on PPP investment is not significant in the private reward mode of government payment, but there is a significant positive effect in the mode of user payment and feasibly insufficient subsidies. This result is consistent with the analysis above, showing that the number of PPP projects related to user payment and feasibly insufficient subsidies increase with the fiscal pressure of the government.

Table 8. Reward Mechanism Test.

Variables	Government Payment	Feasibly Subsidy	Insufficient	User	Payment
<i>deficit</i>	0.928 (1.104)	1.453 * (0.826)		2.018 * (1.131)	
Control variables	Control	Control		Control	
Constants	3.098 (37.12)	74.25 *** (19.90)		74.13 *** (25.31)	
City fixed effect	Control	Control		Control	
Observations	1746	1746		1746	
R ²	0.058	0.081		0.166	

Note: standard errors at the prefectural level are in parentheses. * $p < 0.1$, *** $p < 0.01$.

6. Discussion and Conclusions

6.1. Discussion

In recent years, the adoption of the PPP policy for financing by various local governments in the infrastructure sector has received increasing attention from scholars. This paper examines the impact of fiscal pressure on PPP investments using the data from local governments in China from 2000 to 2015. Table 4 shows that fiscal pressure has an impact on PPP investment at the local level, based on the regression seen for the sample of 291 prefecture-level cities in China, which confirms previous studies by Hammami et al. (2006) [1], Albalade et al. (2015) [19], and Buso et al. (2017) [36]. In other words, fiscal pressure on local governments can be alleviated through the adoption of PPP investment.

Although the PPP policy has been vigorously promoted, there are few studies on the factors influencing the PPP policy. This paper explores the mechanism influencing local governments to choose the PPP model for financing from the perspective of the marketization process and the structural analysis of regional and industry attributes, which broadens the research perspective of the existing literature and is a great supplement to the research on the influencing factors of the PPP policy. The results are consistent with the studies of Shen et al. (2020) [3] and Feng et al. (2020) [3], which show that the presence of government fiscal pressure promotes the advancement of regional marketization and thus PPP project investment. In particular, the advancement of the marketization process can influence the change in government perceptions, policy, and regulation development, and reduce the inefficiency of cooperation due to information asymmetry [23].

In addition, given the differences between verified regions in China, there are also differences in fiscal pressure and PPP project investment in each region [3]. PPP projects in China are mainly issued in the central region, which has been developing fast and has had a high development potential in recent years and can thus relieve the government's financial pressure by implementing PPP projects. Meanwhile, in the eastern region, the economic development is faster and the urban infrastructure construction is better, so the impact of financial pressure on PPP project investment is not significant. In contrast, the western region is less attractive to social capital because of the high cost of infrastructure construction and the weakness of the government itself, and it is affected by the geographical environment, thus restricting the development of PPP projects [4,5]. According to the characteristics of cities, our paper also divided the cities into two categories, central cities (including provincial capitals and municipalities) and ordinary cities. Because provincial capitals and municipalities directly under the central government have good economic development and high fiscal revenues, while ordinary cities have limited economic development, they need to alleviate the fiscal pressure on local governments with the help of PPP projects, which confirms the previous study by Shen et al. (2020) [3].

Finally, the findings of this paper have important implications for the motivation of local governments to adopt the PPP model for financing in the infrastructure sector and for which industries are more suitable for the PPP policy. If local governments are under

fiscal pressure, different conclusions are generated for PPP projects with different return mechanisms [13]. When fiscal pressure on the government increases, it will reduce the use of government-payment-type projects and increase the use of feasibility gap subsidies and user-payment-type PPP projects considering the subsequent fiscal expenditure, which is very consistent with the logical analysis of this paper.

6.2. Conclusions

Following the downturn in the global economy, the fiscal pressure on the government has increased, hindering the construction of infrastructure and public services. Moreover, in recent years, PPP policy has begun to perform an important role in infrastructure construction and public services. Hence, our paper focuses on researching the relationship between fiscal pressure and PPP investment. The results can be summarized as follows. Firstly, the fiscal pressure on governments has a significant positive effect on PPP investment at the prefectural level. Secondly, the marketization process is the mediated effect of the relationship between fiscal pressure and PPP investment. In other words, fiscal pressure will stimulate the regional marketization process, thus promoting PPP investment. Thirdly, fiscal pressure has a significant positive effect on PPP investment in the middle region, while the effect is not significant in the eastern and western regions. Sequentially, the effect is not significant in central cities, but there is a significant positive effect in ordinary cities. Finally, the effect of fiscal pressure on PPP investment is not significant in the private reward mode of government payment, but there is a significant positive effect in the mode of user payment and feasibly insufficient subsidies.

Our findings have the following policy implications. Local governments and private companies are both crucial participants in PPP projects, and their behavior has a large effect on the success and effectiveness of PPP projects. On the one hand, governments need to convert their mode of intervention into a service-oriented one to promote the development of the economy. Specifically, the government should increase the proportion of private capital in the PPP projects and improve the supply efficiency of public goods. On the other hand, it advances the development of private capital in infrastructure construction and continues to deeply promote market-oriented reform. As the marketization process varies greatly between regions, it is necessary to further narrow the differences between the regions to realize the free flow of resources, factors, and capital. In China, compared with the middle and western regions, the eastern regions have a high level of marketization development. Therefore, it is a challenge to facilitate even development nationwide.

This study has two main limitations that should be noted. First, it covers a small number of data in China over a relatively short period. Future studies should include a greater number of locations and a longer evaluation period. Second, we only provide a macrolevel analysis on regional fiscal pressure and PPP investments in China and further analyze the effect from other countries worldwide. Future studies could consider the distinction between the two.

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